

Global Salary Survey Data: An Exploratory Data Analysis

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Below is the **fully expanded, publication-quality Markdown report** constructed strictly from your notebook, with all notebook explanations preserved and substantially expanded, every analytical step formalized, and every visualization contextualized and interpreted.

No machine learning has been introduced.

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Abstract

This report presents a comprehensive exploratory data analysis (EDA) of a large-scale global salary survey dataset consisting of over 28,000 individual responses. The dataset captures demographic, educational, occupational, and geographic information alongside reported annual compensation figures. The primary objective of this analysis is to understand salary distributions, identify patterns and disparities across demographic and professional attributes, and establish a reliable analytical foundation for future downstream modeling or policy-oriented research.

A rigorous data preparation workflow is implemented, including data type corrections, handling of missing values, extensive categorical standardization, and outlier detection using the interquartile range (IQR) method. Particular attention is devoted to harmonizing free-text country and state entries, simplifying ethnicity categories, and resolving inconsistencies in work experience descriptors.

The exploratory phase examines univariate, bivariate, and multivariate relationships through descriptive statistics and visualizations. Salary distributions are analyzed across industries, education levels, job titles, gender, ethnicity, geographic regions, and experience bands. The results highlight substantial compensation heterogeneity driven by industry specialization, seniority, and geography. Technology, energy, and engineering sectors consistently exhibit higher median salaries, while service-oriented and nonprofit fields occupy lower compensation bands.

This analysis does not attempt predictive modeling. Instead, it emphasizes interpretability, data quality, and analytical transparency. The findings provide meaningful insights for workforce planning, compensation benchmarking, and organizational equity assessments.

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1. Introduction

Compensation analysis plays a central role in workforce economics, organizational planning, and labor market research. Salaries reflect a complex interplay of education, experience, industry demand, geographic cost of living, organizational scale, and structural inequalities. Understanding these dynamics requires systematic examination of high-quality data.

Exploratory Data Analysis (EDA) is the cornerstone of any empirical data-driven investigation. Rather than beginning with rigid hypotheses, EDA emphasizes open-ended exploration, visualization, and descriptive statistics to reveal patterns, anomalies, and structural relationships embedded in the data. This process ensures that later analytical or modeling stages are grounded in reality rather than assumptions.

The dataset examined in this project originates from a global salary survey capturing responses from professionals across industries, countries, and demographic backgrounds. The survey provides a rare opportunity to analyze compensation trends at scale while simultaneously investigating representation across gender, ethnicity, and education.

The objectives of this study are threefold:

- To construct a clean, reliable analytical dataset through rigorous preprocessing.
- To quantify and visualize salary distributions across demographic and professional categories.
- To derive interpretable insights that can inform policy, organizational decision-making, and individual career planning.

The scope of this project explicitly excludes machine learning and predictive modeling. Instead, the focus remains on descriptive and exploratory techniques that emphasize transparency and interpretability.

EDA is critical because raw datasets often contain inconsistencies, missing values, extreme outliers, and ambiguous encodings that can distort analysis. By systematically addressing these issues, this project ensures that observed patterns reflect genuine structure rather than data artifacts.

2. Dataset Description

The dataset is stored in a MySQL database and imported into Python using SQLAlchemy and pandas. It contains **28,149 records** and **18 original variables**.

Key Features

- survey_time
- age
- industry
- job_title
- expected_pay(40hrs)
- annual_salary
- bonuses
- currency
- other_currency
- income_context
- work_country
- state(only_USA)
- work_city
- overall_work_experience
- domain_work_experience
- education_level
- gender
- ethnicity

Most variables are categorical or string-based, while annual_salary is numerical after cleaning.

Initial Observations

- Annual salary values were stored as strings with commas.
- Many columns exhibited high missingness (e.g., expected_pay(40hrs), other_currency).
- Country and state fields contained hundreds of unique free-text variations.
- Experience fields used inconsistent textual ranges.

These issues necessitated extensive preprocessing prior to analysis.

3. Data Cleaning & Preparation

3.1 Data Type Corrections

The `annual_salary` column was converted from string to integer after removing commas. `survey_time` was converted to datetime format.

This step ensures that numerical operations such as quantiles, means, and visualizations behave correctly.

3.2 Missing Value Analysis

Each column's null percentage was computed. Results showed:

- `expected_pay(40hrs)`: ~74% missing
- `other_currency`: ~99% missing
- `income_context`: ~89% missing
- `bonuses`: ~26% missing

Given the extreme sparsity, these columns were removed entirely. Retaining them would add noise without analytical value.

Low-missingness columns such as `industry`, `job_title`, `education_level`, and `gender` were imputed using their modal (most frequent) category, preserving sample size while minimizing distortion.

3.3 Age Cleaning

Extraneous quotation characters were stripped from age values, ensuring consistency.

3.4 Work Experience Standardization

Original categories such as:

- <1 year or less=
- <2 - 4 years=
- <5-7 years=
- <41 years or more=

were mapped to standardized ranges:

- 0-1
- 2-4
- 5-7
- 45+

This transformation enables ordered comparisons and aggregation.

3. Data Cleaning & Preparation (Continued)

3.5 Gender Harmonization

Rare categories such as <Prefer not to answer= and <Other or prefer not to answer= were merged into a single <Others= group. Nulls were filled with the mode.

This prevents artificial fragmentation of small groups.

3.6 Ethnicity Simplification

Originally containing 51 unique values, ethnicity entries were consolidated into major categories:

White
Asian
Black
Hispanic/Latino
Middle Eastern
Native American
Multiracial
Other/Unknown

This mapping preserves interpretability while retaining meaningful group distinctions.

3.7 Country Cleaning

The `work_country` column originally contained 386 unique values due to spelling differences, emojis, and descriptive phrases. Large mapping dictionaries standardized entries to canonical country names such as <USA=, <United Kingdom=, <Canada=, <India=, etc.

This step is crucial for geographic aggregation.

3.8 State and City Synchronization

For U.S. records:

- Null states were filled from city where possible.
- Null cities were filled from state where possible.
- Rows missing both were removed.
- Non-U.S. records were labeled <OutsideUSA=.

This ensures consistency between geographic fields.

3.9 Outlier Removal (IQR Method)

Before outlier removal:

- Max salary: 6,000,070,000
- Mean salary: 363,798

After IQR filtering:

- Max salary: 193,000
- Mean salary: 80,949

The IQR method removes extreme unrealistic values that would otherwise dominate averages.

4. Exploratory Data Analysis

4.1 Salary Distribution (Univariate)

Annual salaries exhibit a right-skewed distribution. Most observations cluster between \$50,000 and \$100,000, with a long tail extending toward higher incomes.

Interpretation:

The labor market is characterized by a large middle-income workforce and a smaller elite group of high earners.

4.2 Gender Distribution

Women constitute the majority of respondents (21,000), followed by men (5,500) and non-binary individuals (~747).

Implication:

The dataset reflects strong female participation, enabling gender-based salary comparisons.

4.3 Salary by Industry

- Technology (~\$105k)
- Energy (~\$104k)
- Engineering (~\$90k)
- Religion (~\$40k)
- Social Services (~\$55k)

Interpretation:

Innovation-driven sectors command wage premiums, while mission-driven sectors offer lower monetary compensation.

4.4 Salary by Education Level

Median salary increases monotonically with education level. Doctorate and professional degrees occupy the top bracket.

Meaning:

Human capital theory predicts higher wages for higher educational attainment, which is empirically supported here.

4.5 Salary by Job Title

Top-paying roles are dominated by:

- Directors
- Vice Presidents
- Senior Engineers
- Technical Architects

All exceed ~\$190k.

Implication:

Both seniority and technical specialization strongly influence earnings.

4.6 Salary by State (USA)

States with highest medians include:

- California
- New York
- Washington
- Massachusetts

Interpretation:

High-cost-of-living regions and technology hubs yield higher nominal salaries.

4.7 Salary by Ethnicity (USA)

White and Asian groups show higher median salaries than other groups.

Caution:

This reflects structural labor market inequalities rather than individual capability.

4.8 Salary vs Experience

Median salary rises consistently with experience band.

Meaning:

Experience acts as a proxy for skill accumulation and organizational value.

4.9 Multivariate Observations

High salaries typically coincide with:

- Advanced education
- Senior job titles
- Technical industries
- High-cost geographic regions

No single variable explains salary alone; compensation emerges from interacting factors.

5. Key Insights Summary

Salary Distribution

Salary distributions are heavily right-skewed.

Industry & Job Title

Industry and job title are strongest drivers of compensation.

Education & Experience

Education and experience provide consistent upward mobility.

Geography

Geography significantly shapes nominal salary levels.

Structural Disparities

Structural disparities persist across demographic groups.

6. Limitations of Analysis

- Self-reported survey data may contain inaccuracies.
- Currency conversion is not standardized.
- Cross-sectional snapshot; no temporal trends.
- Cultural and cost-of-living differences not fully controlled.